EA

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LESSON - Setting up Tesseract OCR

Installation

Note: This will also install Leptonica, the image processing library used by Tesseract

Windows

- Download and install from
  Make sure “Add Tesseract to the system PATH” is selected
  https://github.com/UB-Mannheim/tesseract/wiki (https://github.com/UB-Mannheim/tesseract/wiki)

Mac

- Run command
  `brew install tesseract`

Download fast model

Windows

- Locate tessdata folder. Go to that directory in the command prompt
  `cd C:\Program Files\Tesseract-OCR\tessdata`
- Download the fast model data
  `wget -O eng_fast.traineddata https://github.com/tesseract-ocr/tessdata_fast/raw/main/eng.traineddata`

Mac

- Locate tessdata folder. Go to that directory in the terminal
  `cd /opt/homebrew/share/tessdata`
- Download the fast model data
  `curl -L -o eng_fast.traineddata https://github.com/tesseract-ocr/tessdata_fast/raw/main/eng.traineddata`

Add Tesseract to CMakeLists.txt

Windows

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Add Tesseract to CMakeLists.txt

Windows

- Example

```
include_directories(
    "C:/Program Files/Tesseract-OCR/include"
)

link_directories(
    "C:/Program Files/Tesseract-OCR/lib"
)

target_link_libraries(ocr_demo PRIVATE tesseract)
```

Mac

- Example

```
include_directories(
    /opt/homebrew/Cellar/leptonica/1.86.0/include
    /opt/homebrew/Cellar/tesseract/5.5.1/include
)

link_directories(
    /opt/homebrew/Cellar/leptonica/1.86.0/lib
    /opt/homebrew/Cellar/tesseract/5.5.1/lib
)

target_link_libraries(ocr_demo PRIVATE tesseract leptonica)
```

Sample Code

CMakeLists.txt (https://dlsu.instructure.com/courses/224088/files/28931030?wrap=1) ↓
(https://dlsu.instructure.com/courses/224088/files/28931030/download?download_frd=1)
ocr_demo.cpp (https://dlsu.instructure.com/courses/224088/files/28933318?wrap=1) ↓

Windows

- Example

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(https://dlsu.instructure.com/courses/224088/files/28933318/download?download_frd=1)
image.png (https://dlsu.instructure.com/courses/224088/files/28931037)

- Build and run

```
mkdir build
cd build
cmake ..
cmake --build .
```

- For additional details about Leptonica image operations
  https://tpgit.github.io/UnOfficialLeptDocs/leptonica/operations.html
  (https://tpgit.github.io/UnOfficialLeptDocs/leptonica/operations.html)